Student ID:	Seat number:

Semester 2 Exams, 2026

School of Mathematics and Statistics

MAST30027_26_SM2 – Modern Applied Statistics

MOCK PAPER

Reading time: 15 minutes

Writing time: 3 hours

Permitted materials

- Pencils, pens, erasers, highlighters.
- Calculator: Casio FX-82 (any suffix), non-programmable.
- Student notes: 1 × double-sided A4 page of notes.

Exam instructions

- This paper contains **9 questions worth 78 marks**. Marks for each part are shown in square brackets.
- During reading time you may not write, type, or select answers.
- Show all working. Answers without supporting calculations or explanation may receive no marks.
- Give numerical answers to **2 decimal places** unless the question says otherwise.
- You will not be asked to write R code. R code and output are shown only so that you can read, interpret and compute from them. The output is real R output.
- If you need more space, use the pages at the back of the paper and note this in the answer box.
- The probability density and mass functions of the distributions on the next page are not repeated in the questions. Densities of other distributions (Binomial, Beta, etc.) are given only where a question needs them.

Question	1	2	3	4	5	6	7	8	9	Total
Marks available	7	8	9	10	7	10	17	5	5	78
Marks awarded										

Useful distributions

- **Normal**(mean = μ, variance = σ^2). The Normal distribution with mean μ and variance σ^2 has the probability density function

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

  and has the expectation $\mathbb{E}(X) = \mu$.

- **Exponential**(rate = λ). The exponential distribution with rate $\lambda > 0$ has the probability density function

$$f(x; \lambda) = \lambda e^{-\lambda x}$$

  for $x \geq 0$, and has the expectation $\mathbb{E}(X) = \frac{1}{\lambda}$.

- **Gamma**(shape = a, rate = b). The Gamma distribution with shape $a > 0$ and rate $b > 0$ has the probability density function

$$f(x; a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$$

  for $x > 0$, and has the expectation $\mathbb{E}(X) = \frac{a}{b}$.

- **Inverse-Gamma**(shape = a, scale = b). The Inverse-Gamma distribution with shape $a > 0$ and scale $b > 0$ has the probability density function

$$f(x; a, b) = \frac{b^a}{\Gamma(a)} x^{-a-1} e^{-b/x}$$

  for $x > 0$, and has the expectation $\mathbb{E}(X) = \frac{b}{a-1}$ for $a > 1$.

- **Poisson**(rate = λ). The Poisson distribution with rate $\lambda > 0$ has the probability mass function

$$f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

  for $x = 0, 1, \ldots$, and has the expectation $\mathbb{E}(X) = \lambda$.

- **Categorical**(π_1, π_2). The categorical distribution with event probabilities π_1 and π_2, where $\pi_1 > 0$, $\pi_2 > 0$ and $\pi_1 + \pi_2 = 1$, has the probability mass function

$$f(x; \pi_1, \pi_2) = \pi_x$$

  for $x = 1, 2$, and has the expectation $\mathbb{E}(X) = 2 - \pi_1$.

- **Categorical**(π_1, π_2, π_3). The categorical distribution with event probabilities π_1, π_2 and π_3, where $\pi_1 > 0$, $\pi_2 > 0$, $\pi_3 > 0$ and $\pi_1 + \pi_2 + \pi_3 = 1$, has the probability mass function

$$f(x; \pi_1, \pi_2, \pi_3) = \pi_x$$

  for $x = 1, 2, 3$.

Question 1 (7 marks)

A gene has two alleles, A and a. A proportion $\theta \in (0, 1)$ of all alleles in a large population are of type A. Under random mating, an individual has genotype AA, Aa or aa with probabilities θ^2, $2\theta(1-\theta)$ and $(1-\theta)^2$ respectively.

Let $X_1, \ldots, X_n$ be the genotypes of n independent individuals, coded $1 = AA$, $2 = Aa$, $3 = aa$, so that

$$X_i \sim \text{Categorical}\big(\theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big), \qquad i = 1, \ldots, n.$$

Let N_k be the number of individuals with genotype k ($k = 1, 2, 3$), so $N_1 + N_2 + N_3 = n$.

(a) What is the log-likelihood of θ? Write it in terms of N_1, N_2, N_3. [2 marks]

(b) What is the Fisher information $I(\theta)$ (for the whole sample)? You may use $\mathbb{E}(N_1) = n\theta^2$, $\mathbb{E}(N_2) = 2n\theta(1-\theta)$ and $\mathbb{E}(N_3) = n(1-\theta)^2$. Simplify your answer. [3 marks]

(c) Find the MLE of θ and its asymptotic distribution. Then, for $n = 100$ with $(N_1, N_2, N_3) = (34, 46, 20)$, compute the MLE and the estimated variance of its asymptotic distribution (give the variance to 4 decimal places). [2 marks]

Question 2 (8 marks)

In a survey, the number N of insects caught in a trap that is left out for m days is modelled as $N \sim \text{Poisson}(m\mu)$, where $\mu > 0$ is the mean number caught per day and $m > 0$ is known. We record the daily catch rate $Y = N/m$, which takes values in $\{0, 1/m, 2/m, \dots\}$.

(a) Show that the probability mass function of Y can be written in the exponential dispersion family form

$$f(y; \theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}.$$

Identify θ, ϕ, $a(\phi)$, $b(\theta)$ and $c(y, \phi)$. Take $\phi = 1/m$. [4 marks]

(b) What is the canonical link function? Using $b(\theta)$, derive $\mathbb{E}(Y)$ and $\text{Var}(Y)$, and give the variance function $v(\mu)$. [2 marks]

(c) Now suppose $Y_i = N_i/m_i$ for $i = 1, \dots, n$ with different known m_i, and a GLM with the canonical link is fitted by IWLS. The working response z_i has $\text{Var}(z_i) = g'(\mu_i)^2 \, v(\mu_i) \, a_i$, where $a_i = a(\phi)$ evaluated at m_i. Find the IWLS weight $W_{ii} = 1/\text{Var}(z_i)$ in terms of m_i and μ_i, and explain in one sentence why a trap left out for more days receives more weight. [2 marks]

Question 3 (9 marks)

A researcher studies fungal infection in 16 orchard plots. In each plot 30 trees were inspected and `inf` of them were infected (`ok` $= 30 - $ `inf`). For each plot we have `rain`, the mean monthly rainfall (cm), and `spray`, a factor with levels `none` (8 plots, the baseline) and `sprayed` (8 plots). Let p be the probability that a tree is infected. Study the R code and output below, then answer the questions.

```
> fit1 <- glm(cbind(inf, ok) ~ rain + spray, family = binomial, data = orch)

> fit0 <- glm(cbind(inf, ok) ~ rain, family = binomial, data = orch)

> summary(fit1)

Call:
glm(formula = cbind(inf, ok) ~ rain + spray, family = binomial,
    data = orch)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.67973    0.36348  -4.621 3.81e-06 ***
rain          0.29876    0.05337   5.597 2.18e-08 ***
spraysprayed -0.64442    0.21965  -2.934  0.00335 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 48.463  on 15  degrees of freedom
Residual deviance: 14.437  on 13  degrees of freedom
AIC: 80.444

Number of Fisher Scoring iterations: 4


> vcov(fit1)
             (Intercept)         rain spraysprayed
(Intercept)   0.13211643 -0.017971204  0.018175543
rain         -0.01797120  0.002848834 -0.005853401
spraysprayed  0.01817554 -0.005853401  0.048244417

> anova(fit0, fit1, test = "Chisq")
Analysis of Deviance Table

Model 1: cbind(inf, ok) ~ rain
Model 2: cbind(inf, ok) ~ rain + spray
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        14     23.372
2        13     14.437  1   8.9356 0.002797 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Let model 1 be $\text{logit}(p) = \beta_0 + \beta_1 \,\text{rain} + \beta_2 \,\text{spray}_{\text{sprayed}}$.

(a) Compute the estimated odds ratio for spraying (at the same rainfall) and the estimated odds ratio for a one cm increase in rainfall (same spray status). Interpret both in words. [2 marks]

(b) Estimate p for a sprayed plot with `rain` $= 8$, and give an approximate 95% confidence interval for p. Use the normal quantile 1.96, and construct the interval on the linear predictor scale before transforming. [3 marks]

(c) Test whether `spray` is needed in the model once `rain` is included, using a likelihood ratio test at the 5% level. State the hypotheses, the test statistic and its null distribution, and your conclusion. (The 95% quantile of χ^2_1 is 3.84.) [2 marks]

(d) The 95% quantile of χ^2_{13} is 22.36. Is model 1 adequate by the residual deviance? Give one reason why the binomial assumption of independent trees with a common p within a plot could fail in practice, and say what that would do to the standard errors. [2 marks]

Question 4 (10 marks)

Seabirds were counted on 36 boat transects. For each transect the data frame `bird` records the number of birds `count`, the transect length `km`, the mean sea depth `depth` (m), the sea-surface temperature `sst` (°C) and the `zone`, a factor with levels `inshore` (baseline), `shelf` and `slope`. Longer transects see more birds, so the log length is used as an offset. Study the R code and output below.

```
> mA <- glm(count ~ depth + sst + zone + offset(log(km)), family = poisson, data = bird)

> mB <- glm(count ~ depth + offset(log(km)), family = poisson, data = bird)

> summary(mA)

Call:
glm(formula = count ~ depth + sst + zone + offset(log(km)), family = poisson,
    data = bird)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.034161   0.317044   3.262 0.001107 **
depth       -0.006366   0.001689  -3.770 0.000164 ***
sst          0.029598   0.018056   1.639 0.101165
zoneshelf    0.321462   0.096314   3.338 0.000845 ***
zoneslope    0.145690   0.082450   1.767 0.077228 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 245.24  on 35  degrees of freedom
Residual deviance: 222.13  on 31  degrees of freedom
AIC: 403.81

Number of Fisher Scoring iterations: 5


> (phiA <- sum(residuals(mA, type = "pearson")^2 / mA$df.residual))
[1] 7.16976

> (phiB <- sum(residuals(mB, type = "pearson")^2 / mB$df.residual))
[1] 6.747001

> anova(mB, mA, test = "Chisq")
Analysis of Deviance Table

Model 1: count ~ depth + offset(log(km))
Model 2: count ~ depth + sst + zone + offset(log(km))
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        34     234.27
2        31     222.13  3   12.131 0.006947 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

(a) Give two pieces of quantitative evidence from the output that model A suffers from overdispersion. [2 marks]

(b) Model C is model A refitted with `family = quasipoisson`, and model D is model B refitted with `family = quasipoisson`. The command `anova(modelD, modelC, test = "F")` is then run, but its output is hidden. Compute the F-statistic it would report and state its degrees of freedom. The 95% quantile of $F_{3,31}$ is 2.91. What do you conclude about dropping `sst` and `zone`? [3 marks]

(c) Using the output above, compute the standard error of the `depth` coefficient allowing for overdispersion. Hence give a 95% confidence interval for the factor by which the expected bird density (per km) changes for each additional 10 m of depth. Use $t_{31,0.975} = 2.04$. [3 marks]

(d) In 2–3 sentences, explain how the quasi-Poisson model differs from the Poisson model, and why the Chisq analysis of deviance and the F-test disagree about dropping `sst` and `zone`. Which would you trust? [2 marks]

Question 5 (7 marks)

Let $Y_1, \ldots, Y_n$ be independent gamma random variables with means μ_i and variance function $v(\mu) = \mu^2$. We fit the GLM with the canonical (inverse) link,

$$g(\mu_i) = \frac{1}{\mu_i} = \eta_i = \beta_0 + \beta_1 x_i.$$

(a) Find $g'(\mu)$, and show that the IWLS weights $W_{ii} = 1/\big(g'(\mu_i)^2 v(\mu_i)\big)$ simplify to μ_i^2. [1 mark]

(b) A small data set has $x = (0, 1, 2)$ and $y = (2.4, 1.7, 1.1)$. Starting from $\beta^0 = (0.45, 0.10)^T$, compute η_i, μ_i, the working responses z_i and the weights W_{ii} for $i = 1, 2, 3$. Carry at least 4 decimal places. [2 marks]

(c) Write down the weighted least squares formula for β^1. Compute $X^T W X$ and $X^T W z$, and hence the new estimate β^1 (2 d.p.). [2 marks]

The remaining part uses a different data set. A gamma GLM `m` (inverse link, the default) is fitted to the repair cost (`cost`, in \$1000s) of $n = 8$ machines against their age in years. Study the output.

```
> t(d)
     [,1] [,2] [,3] [,4] [,5] [,6] [,7]  [,8]
age  1.00 2.00 3.00 4.00 5.00 6.00 7.00 14.00
cost 1.25 1.91 2.23 1.66 1.95 1.07 0.25  1.05

> m <- glm(cost ~ age, family = Gamma, data = d)

> summary(m)

Call:
glm(formula = cost ~ age, family = Gamma, data = d)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.48605    0.18178   2.674   0.0368 *
age          0.04848    0.03753   1.292   0.2439
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Gamma family taken to be 0.1985256)

    Null deviance: 2.4705  on 7  degrees of freedom
Residual deviance: 2.0854  on 6  degrees of freedom
AIC: 21.448

Number of Fisher Scoring iterations: 5


> round(hatvalues(m), 3)
    1     2     3     4     5     6     7     8
0.414 0.249 0.163 0.128 0.128 0.151 0.188 0.578

> round(residuals(m, type = "pearson"), 3)
     1      2      3      4      5      6      7      8
-0.332  0.114  0.408  0.129  0.421 -0.169 -0.794  0.223
```

(d) Compute Cook's distance for machine 8 (use $p = 2$ and the $\hat{\phi}$ in the output). Machine 7 has the smallest Pearson residual and machine 8 the largest leverage: say what each of these two machines is, and whether either changes the fit much. [2 marks]

Question 6 (10 marks)

Each of $n = 6$ production units yields two independent measurements X_{j1}, X_{j2} of the same quantity (so j indexes units, $j = 1, \ldots, 6$). Each unit is of one of two types, $Z_j \in \{1, 2\}$, and both measurements on a unit come from its type:

$$Z_j \sim \text{Categorical}(\pi, 1 - \pi),$$
$$X_{j1}, X_{j2} \mid Z_j = k \;\sim\; \text{Normal}(\mu_k, 1) \text{ independently}, \quad k = 1, 2.$$

The variance is known to be 1. We want the MLE of $\theta = (\pi, \mu_1, \mu_2)$ using the EM algorithm. Let $\mathbf{X}_j = (X_{j1}, X_{j2})$ and $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_6)$, $\mathbf{Z} = (Z_1, \ldots, Z_6)$. Observed data:

j	1	2	3	4	5	6
x_{j1}	1.5	0.4	1.8	2.0	-0.3	2.4
x_{j2}	2.5	1.1	2.6	1.4	0.6	1.7

Note: if your final answers contain $P(Z_j = k \mid \mathbf{X}_j, \theta^0)$ you need not give a formula for it in parts (a) and (b).

(a) Derive $Q(\theta, \theta^0) = \mathbb{E}_{\mathbf{Z}|\mathbf{X},\theta^0}\left[\log P(\mathbf{X}, \mathbf{Z} \mid \theta)\right]$. [4 marks]

(b) Derive the M-step updates for π and μ_1. [3 marks]

(c) Take $\theta^0 = (\pi^0, \mu_1^0, \mu_2^0) = (0.4, 0, 3)$. Compute $\gamma_{11} = P(Z_1 = 1 \mid \mathbf{X}_1, \theta^0)$. [2 marks]

(d) For the other units the E-step gives $\gamma_{j1} = P(Z_j = 1 \mid \mathbf{X}_j, \theta^0)$ equal to 0.9836, 0.0099, 0.1672, 0.9995 and 0.0240 for $j = 2, \ldots, 6$. Using your γ_{11} from (c), compute the updated $\hat{\pi}$ and $\hat{\mu}_1$. [1 mark]

Question 7 (17 marks)

Model: we observe (x_i, y_i), $i = 1, 2, 3$, with $x = (1, 2, 3)$ and $y = (1.5, 2.0, 3.5)$, and assume that, given β and τ, the y_i are independent and

$$y_i \sim \text{Normal}\Big(\beta x_i,\ \frac{1}{\tau}\Big).$$

Prior: we impose the prior $p(\beta, \tau) = p(\beta \mid \tau)\, p(\tau)$ with

$$\beta \mid \tau \sim \text{Normal}\Big(0,\ \frac{1}{\tau}\Big), \qquad \tau \sim \text{Gamma}(3,\ 2).$$

Note: the probability density functions of the Normal(μ, σ^2) and Gamma(a, b) distributions are given on the second page of the exam.

(a) **Gibbs sampling:** we wish to perform posterior inference using Gibbs sampling.

(i) Derive the conditional distribution $p(\beta \mid \tau, \mathbf{y})$. If it is a known distribution, identify the distribution name and its parameters (as numbers). [3 marks]

(ii) Derive the conditional distribution $p(\tau \mid \beta, \mathbf{y})$. If it is a known distribution, identify the distribution name and its parameters (these may depend on β). [3 marks]

(b) **Metropolis–Hastings algorithm:** we wish to perform posterior inference using the Metropolis–Hastings algorithm. For the current values of the parameters (β_c, τ_c) we propose new values (β_n, τ_n) independently as follows:

$$\tau_n \sim \text{Gamma}\Big(\text{shape} = 2,\ \text{rate} = \frac{2}{\tau_c}\Big),$$
$$\beta_n \sim \text{Normal}\big(\beta_c,\ 0.5^2\big).$$

Compute the acceptance probability when $(\beta_c, \tau_c) = (1,\ 2)$ and $(\beta_n, \tau_n) = (1.2,\ 3)$, using the data above. Show your work. [6 marks]

(c) **Variational inference:** we approximate the posterior by $q(\beta, \tau) = q(\beta)\, q(\tau)$ and use CAVI.

(i) Derive the optimal $q^*(\tau)$. If it is a known distribution, identify the distribution name and its parameters in terms of $\mathbb{E}_q[\beta]$ and $\text{Var}_q[\beta]$. [4 marks]

(ii) Suppose that at some iteration $\mathbb{E}_q[\beta] = 1.1$ and $\mathrm{Var}_q[\beta] = 0.02$. Compute the parameters of $q^*(\tau)$ and $\mathbb{E}_{q^*}[\tau]$. [1 mark]

Question 8 (5 marks)

We want to sample from the density

$$f(x) = x\, e^{-x^2/2}, \qquad x > 0,$$

by rejection sampling with the proposal $g(x) = e^{-x}$, $x > 0$, i.e. Exponential(rate = 1). Consider this R code for one attempt (the constant `M` is to be chosen):

```
x <- rexp(1, rate = 1)
u <- runif(1)
if (u < x * exp(x - x^2 / 2) / M) x else NA
```

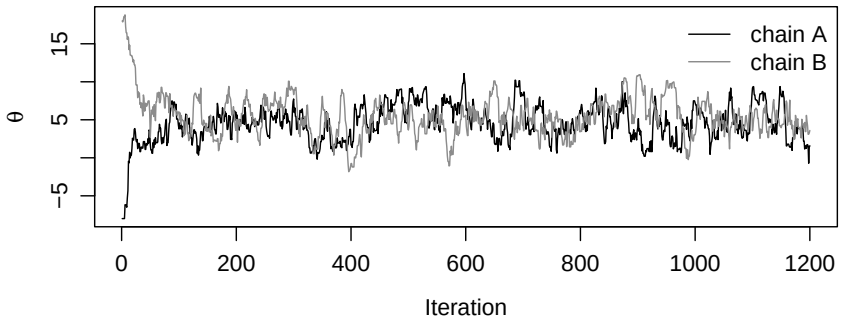
(a) Find the smallest constant M such that $f(x) \le M\, g(x)$ for all $x > 0$. Show your work. [3 marks]

(b) Explain what the last line of the code does. With your M, suppose the proposal is $x = 0.5$ and $u = 0.5$. What does the code return? [1 mark]

(c) What is the probability that a single attempt returns a sample (rather than `NA`)? On average, how many attempts are needed per sample? [1 mark]

Question 9 (5 marks)

A random-walk Metropolis sampler for a parameter θ is run with two chains of 1200 iterations each (chain A starts at -8, chain B at 18). The trace plot and some R output follow. Here `ch` is the 1200×2 matrix of draws (columns = chains).




```
> round(apply(ch[1:300, ], 2, sd), 2)
   A    B
2.65 3.23
> round(sd(as.vector(ch[1:300, ])), 2)
[1] 3.36
> round(apply(ch[301:1200, ], 2, sd), 2)
   A    B
2.29 2.28
> round(sd(as.vector(ch[301:1200, ])), 2)
[1] 2.29
> round(acf(ch[301:1200, 1], plot = FALSE)$acf[2], 2)
[1] 0.92
```

(a) From the trace plot, how many initial iterations would you discard as burn-in? Explain in one or two sentences why burn-in is discarded. [1 mark]

(b) The Brooks–Gelman–Rubin statistic is $\hat{R} = B/W$, where B is the standard deviation of the pooled draws from all chains and W is the average of the within-chain standard deviations; the chains are judged converged if $\hat{R} < 1.05$. Compute $\hat{R}$ for iterations 1–300 and for iterations 301–1200, and say what each tells you. [2 marks]

(c) Use iterations 301–1200 of chain A ($n = 900$ draws, sample standard deviation $s = 2.29$) and the lag-1 autocorrelation in the output. Compute the effective sample size and the Monte Carlo standard error of the chain's sample mean. By what factor is the naive standard error $s/\sqrt{n}$ too small? [2 marks]

Additional working space